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# Unsupervised Classification of Hydrophone Signals With an Improved Mel-Frequency Cepstral Coefficient Based on Measured Data Analysis

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**ABSTRACT** Recently, passive detection technology has developed the ability to detect surface ships based on the noise emissions recorded by hydrophones, making it possible in some cases to classify surface ships. One of the most concerning issues with ships and underwater targets is the current lack of reliable features for unsupervised classification. To solve this problem, this paper proposes an improved Mel-frequency cepstral coefficients (IMFCC) feature for unsupervised clustering of marine targets. As the feature extraction of hydrophone signals rely on preprocessing, the IMFCC adds cyclic modulation spectrum (CMS) and cross-correlation bispectrum (CCB) in the preprocessing module before traditional the Mel-frequency cepstral coefficient process, and the principal component analysis (PCA) is added as the backend processing module. There are four contributions as follows: First, for IMFCC extraction, it combines the advantages of the CMS, CCB, MFCC, and PCA. Second, the two common unsupervised clusters, Gaussian mixture model (GMM) and fuzzy C-means are used to evaluate the CMS, CCB, and several MFCCs—MFCC-vector quantization (MFCC-VQ), MFCC-Gaussian mixture model (MFCC-GMM), TEO-MFCC (Teager Energy Operator based MFCC), and IMFCC. Third, the performances of traditional MFCC, Teager-energy operator (TEO)-MFCC, IMFCC, MFCC-VQ, and MFCC-GMM are discussed under different dimensions, signal-to-noise ratios, distances, and depths. Finally, the experimental results prove that the IMFCC method has a strong anti-interference ability and is robust and has a high-success rate for clustering multitarget or different depth targets, which enables the IMFCC to be a reliable feature for unsupervised classification.

**INDEX TERMS** Cyclic modulation spectrum, CCB, MFCC, GMM, FCM.

## I. INTRODUCTION

The surface and underwater target signals received by the hydrophone are multi-target mixed signals, which are heavily interfered by background noise. To effectively identify all targets, we must accurately classify the targets and detect the number of targets in the signal. For multiobjective unsupervised clustering, these oceanic acoustic signals must be denoised by preprocessing to obtain a certain feature for classification.

The line spectrum features have been studied extensively and are widely used as a basis for classification due to their periodicity [1]–[3]. These features are usually detected by Detection of Envelope Modulation on Noise

(DEMON) and the Cyclic Modulation Spectrum (CMS) [4]. Bicoherence is outstanding in detecting harmonics in underwater acoustic signals and often used for target classification [5], [6]. Mel-frequency cepstral coefficient (MFCC) is a simulated human ear auditory feature and has been widely used in speech recognition [7], [8], e.g. fuzzy-based MFCC used to enhance the difference between the vowel classes [9]; Fractional Fourier Transform (FrFT)-MFCC proposed for music recognition by counter propagation neural network [10]; MFCC-Vector Quantization (MFCC-VQ) and MFCC-Gaussian mixture model (MFCC-GMM) applied for unsupervised voice clustering [11], [12]. However, since the underwater acoustic signals rely on preprocessing, these MFCCs without proper preprocessing steps can hardly obtain high accuracy in unsupervised water target clustering.

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For instance, TEO-MFCC (Teager Energy Operator based MFCC) is simply modified by adding Teager Energy Operator before Mel filtering [13], and binary image-based MFCC is only suited for supervised classifying [14]. A Principal Component Analysis (PCA) is often used to extract the main components of features, as it is efficient in reducing the data dimension with a high correlation level [15], [16]. In regard to unsupervised classification, the most commonly used methods are the Gaussian Mixture Model (GMM) and Fuzzy C-means (FCM). GMM is a model mixed by several Gaussian functions, which has significant performance in the modeling and classification of underwater acoustic signals [17]–[20]. FCM based on k-means has been widely used for target recognition and data classification [21]–[23]. In order to improve clustering accuracy, both of the two methods require better input features, although there is currently no enough effective feature for unsupervised clustering of marine targets.

Keeping this in mind, this paper focus on proposing an improved MFCC (IMFCC) feature for unsupervised clustering ocean targets, which adds CMS and cross-correlation bispectrum (CCB) in the preprocessing module before MFCC, and PCA is added as the backend processing module. The clustering methods used for testing are traditional GMM and FCM. Comparing with the clustering performance of traditional MFCC, MFCC-VQ, MFCC-GMM and TEO-MFCC, IMFCC has the highest accuracy. Although IMFCC takes the longest time in the feature extraction due to the preprocessing step, it costs the least time in the clustering stage, where PCA contributes a lot. This paper is organized as follows: Section II presents the CMS and CCB model, as well as the steps of the IMFCC feature extraction model. Section III discusses the performances of the CMS, bispectrum, traditional MFCC, TEO-MFCC, MFCC-VQ, MFCC-GMM [12] and IMFCC by using GMM and FCM, respectively. Experimental results and comparisons and analysis between the MFCCs and IMFCC are presented in Section IV. Conclusions follow in Section V.

## II. FEATURE EXTRACTION

This section first illustrates the CMS model and the CCB model, respectively. Then the CMS and the CCB are added into the preprocessing stage of IMFCC. Finally, the PCA is introduced as the backend processing module to compress the dimension of IMFCC feature.

The experimental data are collected from the South China Sea. The depth of this area is shown in Fig. 1. The noise used for the study comes from 5 independent vessels and is received by an underwater hydrophone at a depth of 198 m, and all vessels maintain a receiving distance of approximately 10 km with the hydrophone. The SNR is approximately  $-8$  dB, the intercepted data length is 2 minutes, and the sampling frequency is 20000 Hz. These signals are consistent with the raw data used in IV. A.

According to [4], the radiated noise of the ship is included in the noise determination through FFT analysis, which is shown in Fig. 2(a), but not all targets can be distinguished from the power spectral density level.

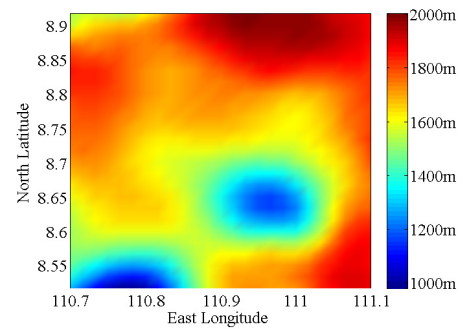


FIGURE 1. The depth of the sea area.

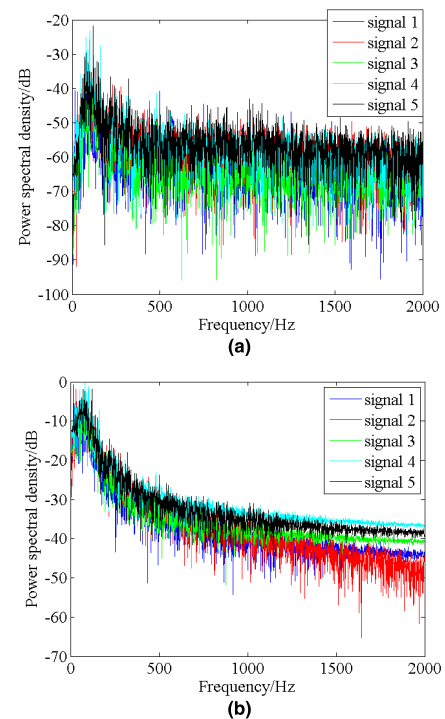


FIGURE 2. Power spectrum of (a) the target radiated noises and (b) the DEMON spectrum.

Compared with Fig. 2(a), the DEMON spectrum in Fig. 2(b) shows that 5 signals exist. However, the effective spectrum information of the DEMON is concentrated in the low frequency, and the DEMON above 1000 Hz is mostly background noise, so it is still impossible to distinguish the 5 signals through DEMON spectrum of the targets.

In Section II, to explain the methods of CMS, CCB and MFCCs and the shapes of the extracted features, it just uses the signal 1 as an example.

### A. CYCLIC MODULATION SPECTRUM FEATURE

The line spectrum is an important feature for judging whether the original signal contains ship radiation noise [3]. Here, we refer to [4] to illustrate the CMS method on extracting line spectrum features. The calculation process is as follows:

Assuming that the hydrophone signal is  $X$ , the sampling signal sequence is  $x(n)$ ,  $n = 0, 1, 2, \dots, N-1$  obtained after sampling at the  $N$ -point. The frequency resolution is  $\Delta f_s = f_s/N$ , the sampling frequency is  $f_s$ , and the sampling period  $T_s = 1/f_s$ .  $x(n)$  is convoluted with the window function  $w(n)$ , where a Hanning window that is  $L$ -long is selected.  $w(n)$  is used to take overlapped snapshots from  $x(n)$ , the number of snapshots is  $L_1 = (N-L)/L_2 + 1$ , and each snapshot of  $x(n)$  has the number  $k = 1, 2, 3, \dots, L_1$ .  $L_2$  is the shifting size equal to the fraction of overlap. Here, the length of  $L_2 = N/L_1$ ,  $f$  is the center frequency with a width  $\Delta f \sim 2\pi/LT_s$ , and  $j^2 = -1$ . Then, the short-time Discrete Fourier Transform (DFT) is calculated over the signal  $x(n)$  in Eq. (1), which returns a narrow-band filtered signal  $x_k(F_L, f)$ , and the framewise sampling frequency is  $F_L = f_s/kL_2$ .

$$x_k(F_L, f) = \frac{1}{L_2} \sum_{n=kL_2}^{kL_2+L-1} x(n)w(n-kL_2)e^{-jnfT_s} \quad (1)$$

DFT is used on the squared magnitude  $|x_k(f_n, f)|^2$  to acquire the time-dependent spectrum, which is calculated to obtain the cyclic frequency  $f_L$  by taking the DFT along each frequency of the spectrogram. This converts  $F_L$  into dual frequency  $f_L$  variables by the square amplitude of the second DFT operation. We calculate the CMS as follows:

$$P(f_L, f) = \frac{1}{L_1} \sum_{k=1}^{L_2} |x_k(F_L, f)|^2 e^{-j\frac{f_L}{F_L}} \quad (2)$$

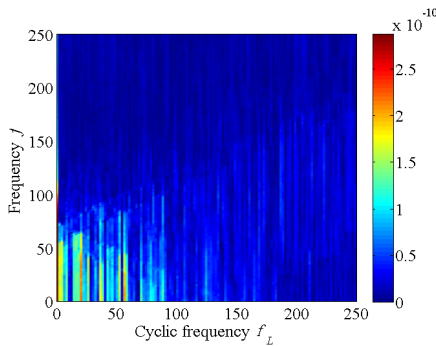


FIGURE 3. Cyclic Modulation Spectrum of signals 1.

Fig. 3 shows the CMS spectrum of signal 1. In this figure, there are strong line spectrum features in the frequency band below 100 Hz, which are the height of lines that can be used for target detection in [4]. Comparing to the signal 1 in Fig. 2(b), CMS can detect clearer line spectrum than DEMON spectrum, which can effectively improve the extraction capacity of line spectrum features of DEMON.

### B. CCB FEATURE

Bispectrum is the most common high-order spectrum analysis (HOSA) method. It is widely used in digital signal processing, including water targets recognition and classification. The quadratic phase coupling (QPC) of bispectrum

is presented with harmonics, which is important for identifying and separating different physical generating mechanisms [4], [5]. Here, to improve the detection performance of the bispectrum on the QPC, we propose acquiring the preprocess signal by performing cross-correlation processing based on the CMS and adopting a bispectrum analysis on it to finally obtain the CCB. Its derivation is computed as follows:

We perform a discrete  $N$ -long IFFT on  $P(f_L, f)$  to obtain signal  $Y(n)$ , where the sampling frequency is also  $f_s$ .  $Y(n)$  is equally divided into  $L_3$  segments. Then, the cross-correlation function  $R_m(\tau)$  is the correlation of every two segments  $y_i(n)$  and  $y_j(n)$ , which is derived as follows:

$$Y(n) = \frac{1}{N} \sum_{f=0}^{N-1} P(f_L, f) e^{j\frac{nf}{f_s}} = \frac{1}{L_3} \sum_{i=1}^{L_3} y_i(n) \quad i = 1, 2, \dots, L_3 \quad (3)$$

$$R_m(\tau) = \frac{1}{L_4} \sum_{n=mL_4}^{(m+1)L_4} y_i(n)y_j(n-\tau) \quad m = 0, 1, \dots, \frac{L_3(L_3-1)}{2} \quad (4)$$

$\tau$  is the time delay of correlation  $R_m(\tau)$ ,  $m$  is the serial number of  $R_m(\tau)$ , and  $L_4 = N/L_3$  is the length of every segment  $y_i(n)$ . The estimated power spectrum over the entire frequency band  $R_m(\tau, f)$  is derived as follows:

$$Q(f) = \frac{2}{L_3(L_3-1)} \sum_{m=1}^{L_3(L_3-1)/2} |R_m(\tau, f)|^2 \quad (5)$$

where  $R_m(\tau, f)$  is the DFT of the  $m$ -th segment correlation  $R_m(\tau)$ , which is used to calculate the discrete bifrequency of the bispectrum  $B(f_1, f_2)$ :

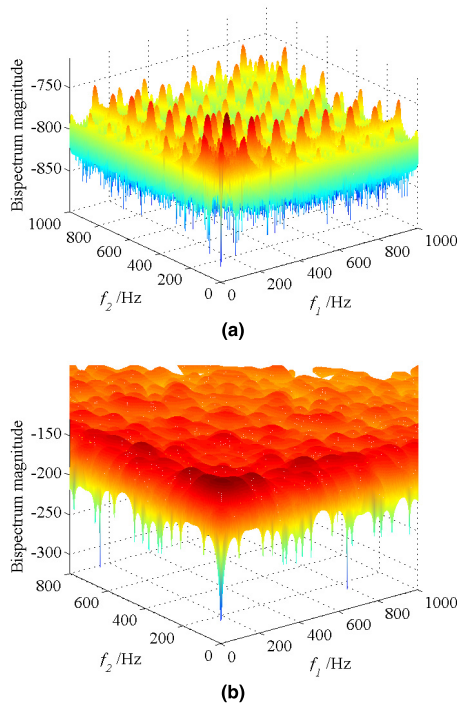
$$B(f_1, f_2) = \frac{2}{L_3(L_3-1)} \sum_{m=1}^{L_3(L_3-1)/2} R_m(\tau, f_1)R_m(\tau, f_2)R_m^*(\tau, f_1+f_2) \quad (6)$$

The CCB  $B(f_1, f_2)$  is actually obtained by using a Fourier transform of the third-order cumulants, where  $(.)^*$  denotes the complex conjugate.

Fig. 4(a) presents the CCB of signal 1. The peaks in the figure are periodic strong phase-coupling harmonics used as target classification features. The phase-coupled harmonics detected by the CCB are partially coincident with the line spectrum detected by the CMS. However, in Fig. 4(b), the phase-coupled harmonics cannot be detected by the conventional bispectrum proposed by reference [5]. The CCB is predicted to be more advantageous than the traditional bispectrum in classification.

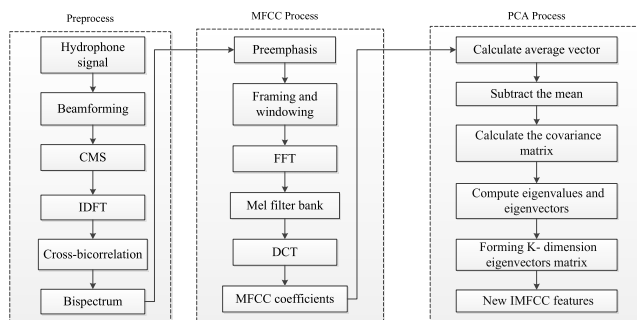
### C. IMPROVED MEL-FREQUENCY CEPSTRAL COEFFICIENTS FEATURE

MFCC is a feature that simulates the human ear receiving acoustic signals and has been used in a variety of acoustic signal processing. When human's ear receives signals, it suppresses noise and channel interference, and performs



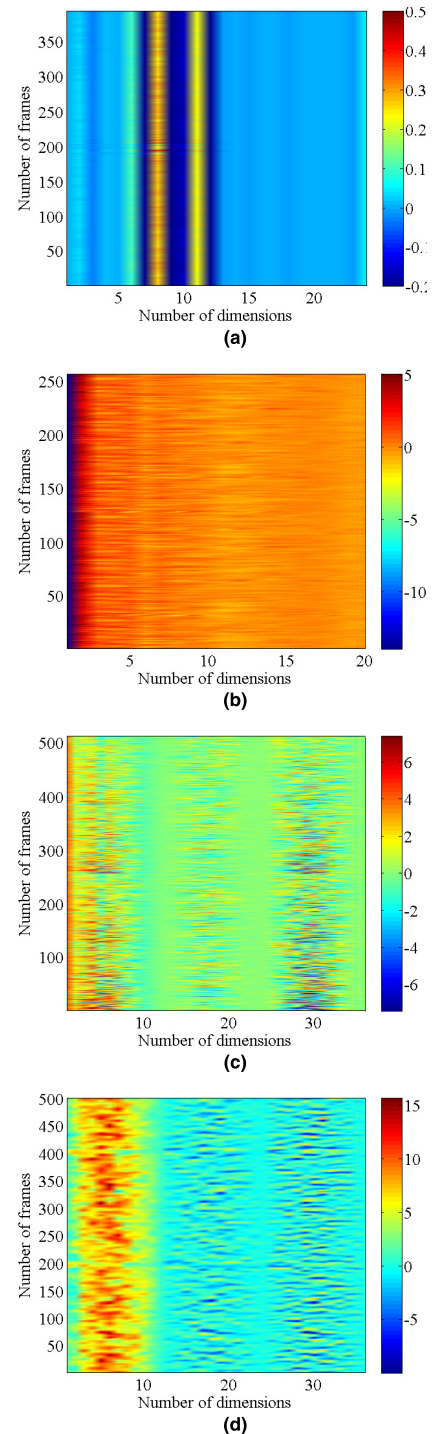
**FIGURE 4.** (a) CCB feature of signal 1 and (b) conventional bispectrum feature of signal 1.

adaptive filtering on the environment to extract a variety of information from the sound signals. Moreover, the inner hair cells of the human ear have the characteristics of half-wave rectification and low-pass filtering, which are equivalent to the CMS envelope extraction. Therefore, this section adds the CMS and the CCB to preprocess the hydrophone signal, and then extracts the MFCC feature with successfully retaining the information of line spectrum and the bispectrum. To reduce the calculation of MFCC, we use PCA to reduce the dimension of IMFCC [15], [16] and to finally obtain the reconstructed IMFCC features shown in Fig. 5.



**FIGURE 5.** IMFCC feature extraction process.

Compared with IMFCC, the traditional MFCC only represents the “MFCC Process” in Fig.5, while the TEO-MFCC just only adds a module named Teager-energy operator (TEO) between FFT and Mel filter bank in the traditional MFCC processing [13]. Similarly, MFCC-VQ only use Vector Quantization (VQ) to compress traditional MFCC without preprocessing, and MFCC-GMM just calculates the GMM



**FIGURE 6.** (a) The 20D IMFCC, (b) 20D MFCC-VQ, (c) the 36D TEO-MFCC and (d) 36D traditional MFCC of signal 1.

result of traditional MFCC by maximum log likelihood function [11], [12]. Nevertheless, for underwater acoustic signals that very rely on preprocessing, these methods without proper preprocessing module can hardly achieve clean results just as shown in Fig. 6.

Fig. 6(a) and 6(b) show the 20-D (dimensional) IMFCC and MFCC-VQ of signal 1, which are subjected to PCA and

VQ compressions, which require 58 s and 61 s in MATLAB, respectively. Fig. 6(c) shows the 36-D TEO-MFCC, costing nearly 12 s, similarly, the traditional MFCC in Fig. 6(d), spends 10 s. Comparing these MFCCs, it can be seen that the IMFCC obtain the cleanest MFCC feature, followed by TEO-MFCC, MFCC-VQ, and finally the traditional MFCC performs the worst, even it takes the least time. In summary, although the computation time will certainly be extended, preprocessing is necessary to extract cleaner IMFCC feature of the hydrophone signal.

### III. CLUSTER ANALYSIS METHOD

This section applies two widely used unsupervised cluster methods, namely, GMM and FCM, to classify the features shown in Section II, and to discuss the performance of these MFCCs and IMFCC.

#### A. CLUSTER ANALYSIS BASED ON GAUSSIAN MIXTURE MODELS

Referring to [17]–[20], GMM is a parametric probability density distribution model. This method obtains the probability of each sample point by assuming that all data points are generated from a mixture of a finite number of Gaussian functions. If the component is sufficiently large, the GMM can approximate any probability density distribution, but easily falls into local extremum. Meantime, GMMs may take up to many hours to calculate.

The GMM considers the data to be combined from  $K$  Gaussian functions, that is, implicitly  $K$  Gaussian functions. Given the data set  $X = \{x_1, x_2, \dots, x_n\}$ ,  $k$  is the number of categories, any GMM distribution is defined as:

$$p(x) = \sum_{k=1}^K \pi_k N(x|\mu_k, \sigma_k) \quad (7)$$

$\pi_k$ ,  $\mu_k$  and  $\sigma_k$  are the weight coefficient, mean and variance of the  $k$ -th Gaussian distribution, respectively. These GMM parameters are solved by an EM algorithm.

*E step:* For the observation point  $x_i$ , the probability  $\gamma(x_i, k)$  generated by the  $k$ -th Gaussian distribution is

$$\gamma(x_i, k) = \pi_k \frac{N(x_i|\mu_k, \sigma_k)}{\sum_{j=1}^K \pi_j N(x_i|\mu_j, \sigma_j)} \quad (8)$$

*M Step:* Find the parameter value corresponding to the maximum likelihood of the distribution

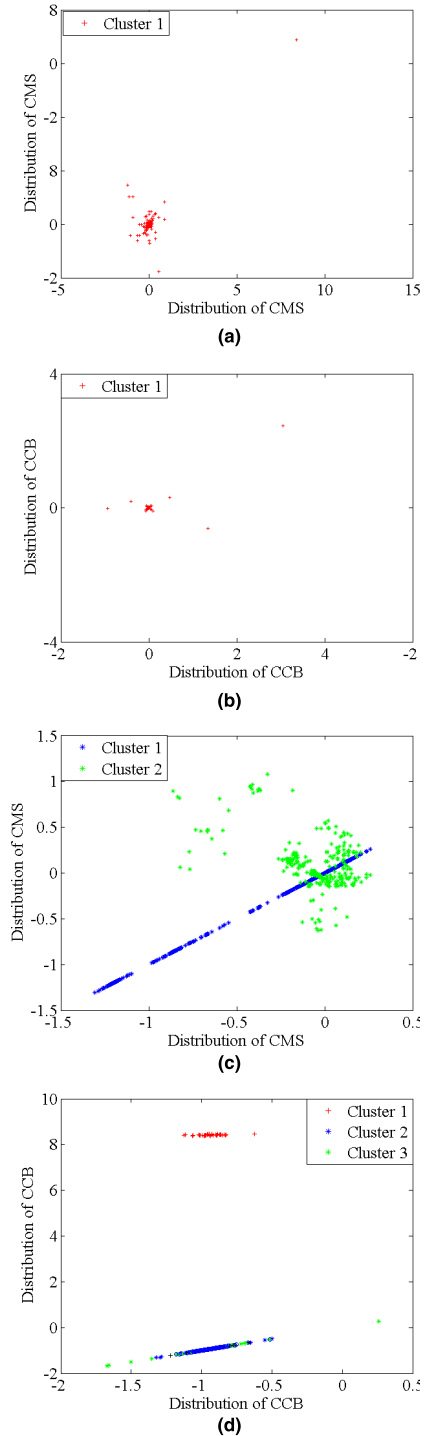
$$\mu_k = \frac{1}{N} \sum_{i=1}^N \gamma(x_i, k) x_i \quad (9)$$

$$\sigma_k = \frac{1}{N_k} \sum_{i=1}^N \gamma(x_i, k) (x_i - \mu_k) (x_i - \mu_k)^T \quad (10)$$

$$N_k = \sum_{i=1}^N \gamma(x_i, k) \quad (11)$$

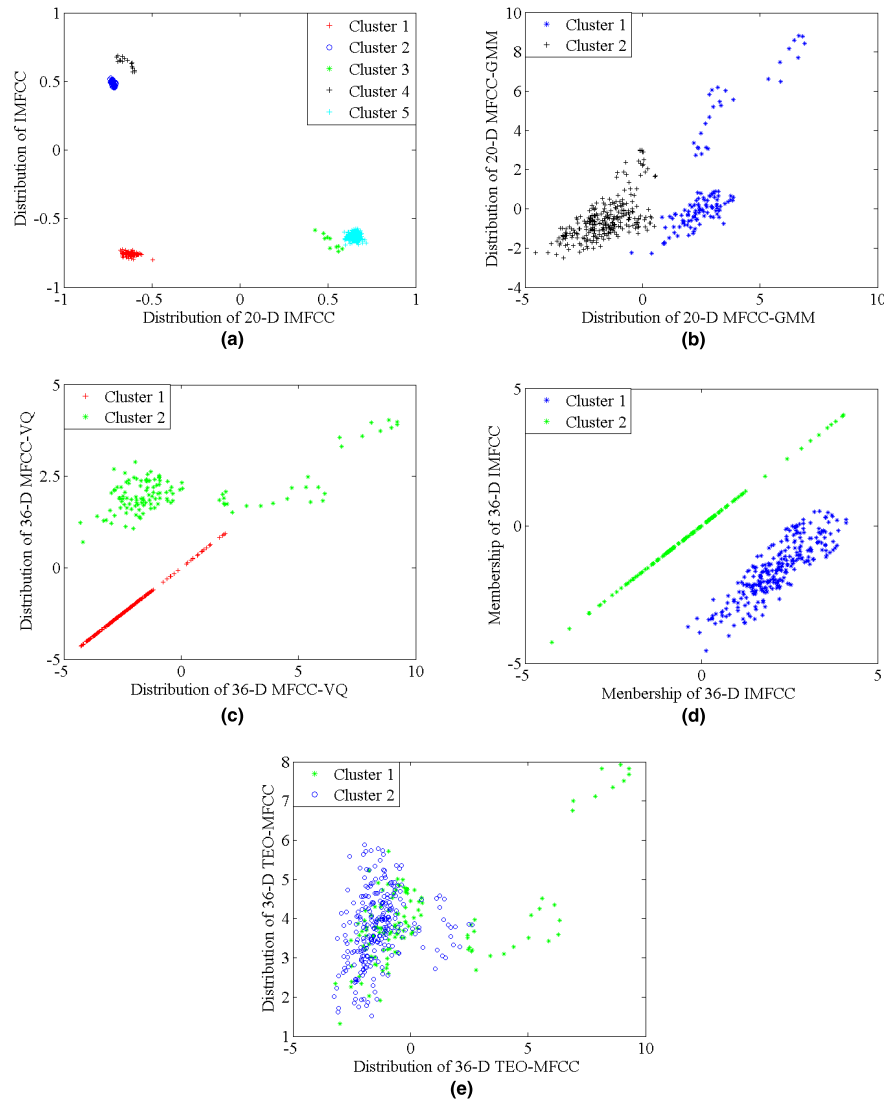
$$\gamma(x_i, k) = N_k / N \quad (12)$$

Update the parameters and repeat the E-step for the next iteration until the algorithm converges.



**FIGURE 7.** GMM results of (a) CMS with 1 cluster, (b) CCB with 1 cluster, (c) CMS with 2 clusters, (d) CCB with 3 clusters.

The Gaussian component  $K$  of the GMM is set to 5, and the initial input parameters of the weight coefficient, mean and variance are all randomly selected. The model is used to classify the features of the 5 signals stated in Section II, with the results shown in Fig. 7 and Fig. 8. Since the solution



**FIGURE 8.** GMM results of (a) 20D IMFCC with PCA, (b) 20D MFCC-GMM (c) 36D MFCC-VQ, (d) 36D IMFCC without PCA, (e) 36D TEO-MFCC.

method of the GMM is based on an EM algorithm, it is possible to fall into local extremum, which is very related to the selection of the initial value.

Fig. 7(a) and (b) show that the GMM cluster analysis cannot classify 5 signals into 5 clusters using the CMS and CCB. Even if the random initial conditions are changed multiple times, the better results of the two features in Fig. 7(c) and 7(d) can only reach up to 2-3 clusters. Therefore, it is difficult for GMM to achieve the desired classification requirements without finding initial input parameters suitable for classifying CMS and CCB.

Given by [7]–[13], except IMFCC, the other MFCCs haven't been preprocessed. According to [12], the performance of MFCC-VQ is weaker than MFCC-GMM in speech classification, while in Fig. 8(b) and 8(c), they show the similar results, only can obtain 2 clusters.

Likely, Fig. 8(d) and 8(e) show that both the IMFCC without PCA processing and the TEO-MFCC also classify 5 signals into 2 clusters. In contrast, the IMFCC features are not prone to falling into a local convergence. Compare Fig. 8(a) and 8(c), the IMFCC and MFCC-VQ are both compressed to 20-D, while IMFCC's result is much better than MFCC-VQ. The main reason is that the noise has been almost filtered out by preprocess in IMFCC, whereas the MFCC-VQ has a larger proportion of noise interference.

The GMM iteration calculation time is relatively long; it takes more than 10 hours to cluster each feature using MATLAB, compared to no more than 1 minute via the FCM method. The more dimensions a feature has, the longer the calculation time is. Therefore, in order to reduce the

calculation time and avoid the clustering results from local convergence, this paper tries to classify five signals via FCM clustering.

### B. FUZZY C-MEANS CLUSTERING

According to [21]–[23], FCM optimizes the objective function to obtain the degree of membership of each sample point to all class centers, thereby determining the cluster of the sample points to classify sample data automatically.

- a) Given the data set  $X = \{x_1, x_2 \cdots x_n\}$ ,  $C$  is the number of categories,  $m_j$  is the cluster center,  $U_j(x_i)$  is the membership function of the  $i$ -th sample corresponding to the  $j$ -class, so the clustering objective function based on the membership function is

$$J = \sum_{j=1}^C \sum_{i=1}^n [U_j(x_i)]^b \|x_i - m_j\|^2 \quad (13)$$

where the degree of membership  $b$  is a weighted index, and  $\sum_{j=1}^C U_j(x_i) = 1$ . To obtain the necessary conditions to minimize the objective function, let

$$\frac{\partial J}{\partial U_j} = \frac{\partial J}{\partial m_j} = 0 \quad (14)$$

$$m_j = \frac{\sum_{i=1}^n [U_j(x_i)]^b x_i}{\sum_{i=1}^n [U_j(x_i)]^b} \quad (15)$$

$$U_j(x_i) = \frac{\|x_i - m_j\|^{-2/(b-1)}}{\sum_{s=1}^C \|x_i - m_s\|^{-2/(b-1)}} \quad (16)$$

Given the number of clustering categories  $C$ , set the iteration convergence conditions, and initialize the membership  $U_j(x_i)$ .

Step (a) is repeated until  $U_j(x_i)$  of each  $x_i$  is stable.

The performance of the FCM method is dependent on the initial clustering center, but it can be updated by iterative calculations to achieve convergence. Fig. 9(a) and 9(b) reflect that even though the FCM can classify the CMS and CCB into five clusters, they are not concentrated together. The IMFCC and MFCC-VQ in Fig. 10(a) and 10(b), both have been compressed to 20-D. The IMFCC with preprocessing module can obtain clear 5 categories, while the result of MFCC-VQ shows 5 mixed clusters. It is similar for Fig. 10(c) and 10(d) that the cluster points of the 36-D IMFCC without PCA and TEO-MFCC appear to be mixed and cannot be identified. Fig. 10(a) presents almost the same results as in Fig. 8(a) obtained by GMM.

Meantime, it costs both IMFCC and MFCC-VQ nearly 10s, followed by IMFCC without PCA, almost 19s, and the maximum of TEO-MFCC, about 21s, used by MATLAB. Compared with GMM, FCM greatly reduces the calculation time and can also obtain good clustering results. Therefore, in terms of saving time, FCM has more advantages than GMM. Thus, in IV, FCM is chosen to cluster signals.

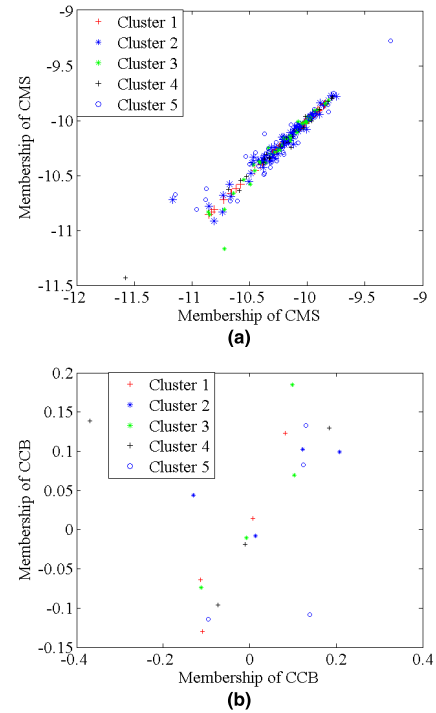


FIGURE 9. FCM results of (a) CMS and (b) CCB.

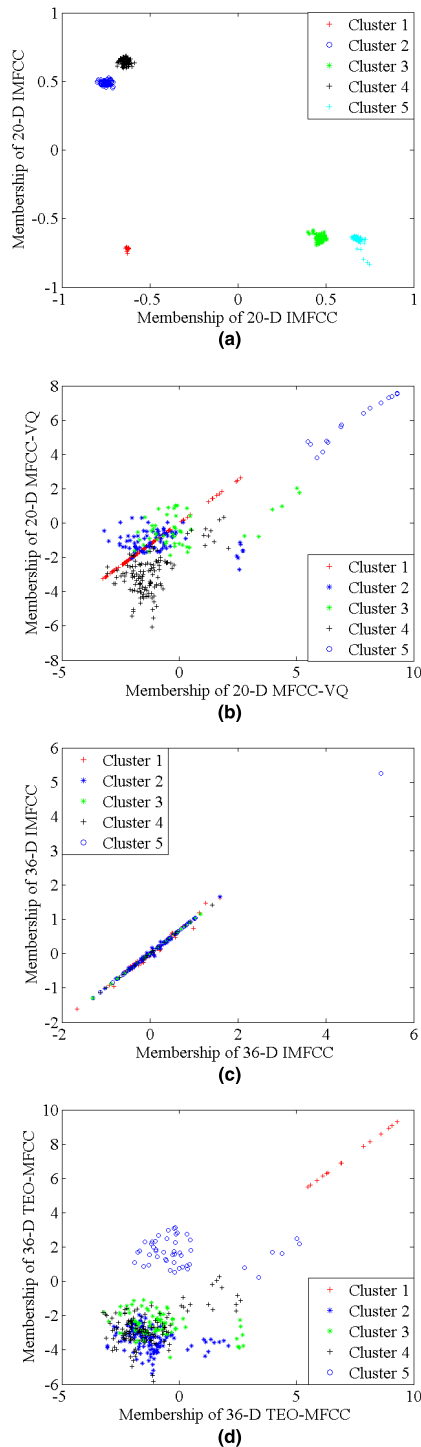
## IV. EXPERIMENTAL RESULTS AND ANALYSIS

### A. COMPARISON AND ANALYSIS BETWEEN IMFCCs and MFCCs

To test the performance of the IMFCC method on the accuracy of the FCM cluster analysis, we use the 5 signals in the Section II. In addition, we change the dimensions of the IMFCC without PCA, the IMFCC with PCA, MFCC-VQ and the TEO-MFCC. Before PCA processing, the MFCC and IMFCC are both 36-D. Then, the traditional MFCC, the IMFCC without PCA and the TEO-MFCC only classify their 1D, 1-4D, 1-8D, 1-12D, 1-16D, 1-24D, 1-28D, 1-32D and 1-36D, and MFCC-VQ and IMFCC with PCA are compressed from 36-D to 1-D, 4-D, 8-D, 12-D, 16-D, 24-D, 28-D, 32-D for testing. When using the FCM, except for the input data, the other input parameters are unchanged, the number of clusters classified is always set to 5, and the variables are only the input data. Table 1 shows the classification results

TABLE 1. The number of clusters of MFCCs in different dimensions.

| MFCC dimensions   | 1 | 4 | 8 | 12 | 16 | 20 | 24 | 28 | 32 | 36 |
|-------------------|---|---|---|----|----|----|----|----|----|----|
| TEO-MFCC          | 2 | 2 | 3 | 2  | 3  | 4  | 4  | 5  | 5  | 5  |
| MFCC-VQ           | 1 | 2 | 3 | 3  | 4  | 5  | 5  | 5  | 3  | 3  |
| Traditional MFCC  | 1 | 1 | 3 | 3  | 3  | 3  | 2  | 2  | 4  | 3  |
| IMFCC without PCA | 4 | 4 | 5 | 5  | 5  | 5  | 5  | 5  | 2  | 3  |
| IMFCC with PCA    | 5 | 5 | 5 | 5  | 5  | 5  | 5  | 5  | 4  | 3  |



**FIGURE 10.** FCM results of (a) 20D IMFCC with PCA, (b) 20D MFCC-VQ, (c) 36D IMFCC without PCA, (d) 36D TEO-MFCC.

of five MFCC features by the FCM. The performance of the traditional MFCC is obviously the worst; it can only classify the 5 targets into no more than 4 clusters.

Table 1 shows that the 1-4 dimensional IMFCC ensures the correct classification with PCA, whereas IMFCC without PCA cannot. It can be seen that the effective information

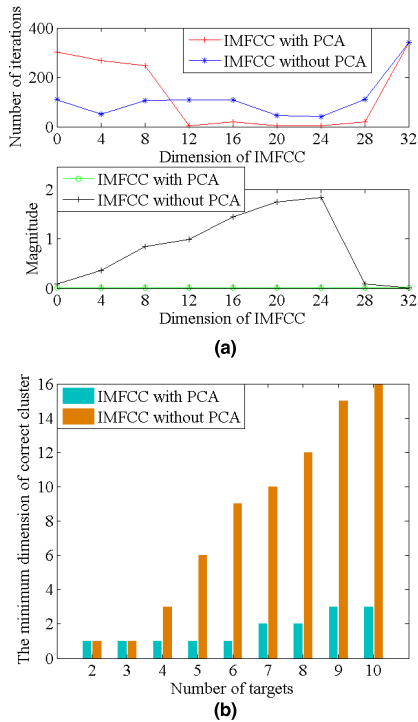
of the IMFCC concentrates more in the low dimensions, and the high-dimensional IMFCC features reducing the FCM accuracy can be compressed to a low dimension by PCA. On the other hand, even MFCC-VQ also compresses the high-dimensional features, due to no pre-processing and the mixed noise interference, it still cannot get accurate clusters except 20D-28D. Similarly, TEO-MFCC can only classify 5 targets when the dimensions are 28D-36D. However, the 28-D MFCC-VQ and TEO-MFCC must spend more time in classification than the 1D IMFCC (23s vs 4s). Therefore, considering the computational cost, the IMFCC with PCA processing has advantages than other TEO-MFCC and MFCC-VQ.

Here, in order to better compare the improvement of the IMFCC by the PCA process, Fig. 11(a) shows the amount of calculation and the magnitude of the objective function  $J$  of the two IMFCCs by the FCM. It reflects that when the dimension is  $> 10$ , the PCA enables the FCM to calculate the IMFCC with very few iterations to obtain a very small objective function. One target signal is randomly selected in 10 different scenarios, 2-10 of the 10 signals are extracted to calculate their IMFCCs (with or without PCA), and the size of the minimum IMFCC required for the FCM is tested for the correct classification. The results are shown in Fig. 11(b). The comparison shows that when the number of targets is  $\leq 6$ , the IMFCC can be correctly clustered even if it is compressed into 1-D features by the PCA. Even if the target is increased to 10, the IMFCC only needs to be compressed into a 3-D feature to identify the correct cluster. In contrast, when the target number is  $\leq 3$ , the FCM can be classified only by the 1-D IMFCC without PCA. However, as the number of targets increases, the minimum dimension of the IMFCC without PCA required for correct clustering also increases dramatically. It can be concluded that the IMFCC processed by the PCA is more advantageous when performing more targets cluster analyses.

## B. ROBUSTNESS ANALYSIS OF MFCCs AND IMFCCs

To test the robustness of these MFCCs, signals added different SNRs are used. Section II mentioned that the SNR of the 5 signals is approximately -8 dB, so the background noise of the 5 signals must be filtered out first. Next, -15, -10, -5, 0, 5, 10, 15, 20, 25 dB Gaussian white noise is added to the 5 denoising signals. Then, referring to the results of Table 1 and Fig. 11, the generated signal is processed separately to obtain the 36D traditional MFCC, the 28D TEO-MFCC, IMFCC without PCA, and the MFCC-VQ and IMFCC with PCA compressed from 36D to 20-D. Then, the FCM clustering is performed using the three MFCCs, the dimensions of MFCCs and IMFCCs remain unchanged, and other input parameters are unchanged.

The results are shown in Table 2. It presents that the 20-D IMFCCs of the 5 signals can be stably classified under different SNRs, and their robustness is significantly better than traditional TEO-MFCC when  $\text{SNR} \leq -5$  dB, followed



**FIGURE 11.** (a) The amount of calculation, objective function and (b) the minimum dimensions required by the correct FCM for different numbers of targets between IMFCCs (with or without PCA).

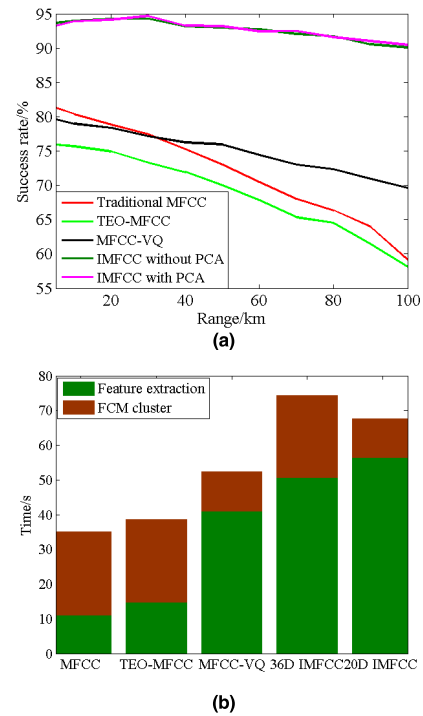
**TABLE 2.** The number of clusters of MFCCs with different SNRs.

| SNR/dB            | -15 | -10 | -5 | 0 | 5 | 10 | 15 | 25 |
|-------------------|-----|-----|----|---|---|----|----|----|
| TEO-MFCC          | 3   | 3   | 4  | 5 | 5 | 5  | 5  | 5  |
| MFCC-VQ           | 1   | 2   | 3  | 3 | 5 | 5  | 5  | 5  |
| Traditional MFCC  | 1   | 2   | 2  | 3 | 3 | 3  | 5  | 5  |
| IMFCC without PCA | 5   | 5   | 5  | 5 | 5 | 5  | 5  | 5  |
| IMFCC with PCA    | 5   | 5   | 5  | 5 | 5 | 5  | 5  | 5  |

by MFCC-VQ when  $\text{SNR} \leq 0$  dB, and traditional TEO-MFCC with  $\text{SNR} \leq -5$  dB.

We selected 186 groups of surface ship data from four targets, where the distance between the target and the hydrophones is 5-100 km, and the SNR decreases with increasing distance. The experimental data is used to test the robustness and time cost of the MFCCs and the IMFCCs. We extract these features with the same dimension as Table 2, the success rate and time spent by MATLAB are shown in Fig. 12.

Fig. 12(a) presents the relationship between distance and success rate of the MFCCs and the IMFCCs, clustered by FCM. It indicates that IMFCCs have lowest decline speed, greater than 90%, and reach to the peak 94.6% at 30 km. Meantime, TEO-MFCC, starting at 79.8% and ending at 70.0%, has a lower drop rate than MFCC-VQ, starting at



**FIGURE 12.** (a) Performance of MFCCs and IMFCCs with the changes of receiving distance and (b) time costed by MFCCs and IMFCCs with distance = 10 km.

82.7% and ending at 59.1%. In contrast, traditional MFCC still occupies the rapidest decline speed, and the lowest success rate, only 57.6% at 100km. The bar chart in Fig. 12(b) shows the time costed by MFCCs and IMFCCs at 10 km, where IMFCC without PCA spends the biggest time, 74.2s, similarly, MFCC-VQ uses 73.97s, followed by IMFCC with PCA, 67.6s. Of course, the time costed by traditional MFCC and TEO-MFCC are 35.1s, 38.5s, nearly half of MFCC-VQ. It is noticing that among them, IMFCCs costs less time than MFCC-VQ in preprocessing stage, and takes the least time in FCM clustering, due to the dimension compression by PCA. Overall, the time-consuming of IMFCC with PCA is medium, while which maintain the highest success rate.

### C. IMFCC CLASSIFICATION OF MIXED SIGNALS BETWEEN UNDERWATER AND SURFACE TARGETS

We launched a simulated submarine noise signal 81 m underwater in the experimental area and selected 95 sets of raw data. The IMFCC characteristics of the simulated submarine noise are shown in Fig. 13. At the same time, 95 sets of experimental ship data were selected at the same receiving distance. These two types of data are divided into 16 groups according to distances of 6, 30, 55, 66 km. The data are processed by the IMFCC method, where the IMFCC of simulated submarine noise is shown in Fig. 13(a), and the FCM result of one group is shown in Fig. 13(b).

The FCM classification is performed by extracting all the IMFCC features of the mixed signals. Statistical classification found that only three groups of data are inaccurately

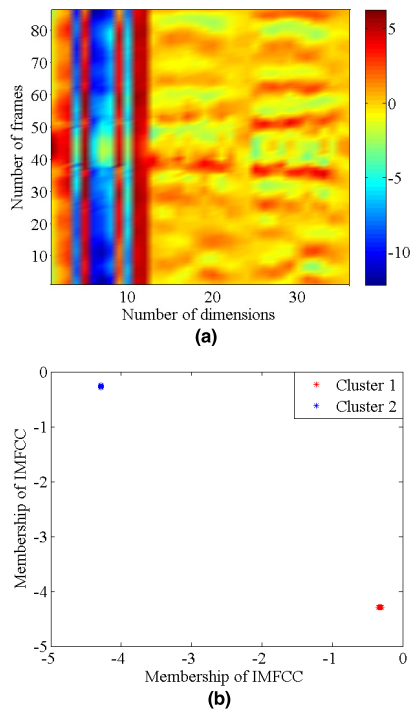


FIGURE 13. (a) IMFCC and (b) FCM of mixed signal.

classified. The experimental results show that the IMFCC method has a high success rate of nearly 98.3% for multitarget classification at different depths.

## V. SUMMARY

This study combines the CMS, CCB and PCA algorithm to improve the MFCC method and apply it to ship acoustic classification. Under unsupervised conditions, the clustering performance of signals from five surface targets was analyzed using the GMM and FCM methods. The results show that IMFCC is the most stable classification feature compared to the single CCB, CMS, traditional MFCC MFCC-VQ, MFCC-GMM and TEO-MFCC. Moreover, even IMFCC required extra time on preprocessing, it cost the smallest time on FCM cluster.

We further discuss the impact of the PCA on the classification performance of IMFCC by changing dimensions, the number of classification targets. Compared to traditional MFCC, TEO-MFCC, MFCC-VQ and IMFCC without PCA, IMFCC with PCA processing has better multitarget (>5) classification ability. The robustness and time cost have been quantized by experimental data, which is proven that IMFCC with PCA has the biggest robustness, while costs nearly 1 minute. To further validate the classification performance of the IMFCC, FCM clustering was performed using both surface and underwater data. The results show that IMFCC is effective for the classification of surface targets and underwater targets. To conclusion, the IMFCC greatly improves the unsupervised classification accuracy of water targets, and maintains a high success rate in experimental test.

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